

--TASK C: Flink Real-Time Incident Detection & Spark Urban Analytics Engine --

Problem Statements — Part I: Apache Flink (Incident Detection)

AQI Emergency:

"When any sensor reports AQI > 300 (Hazardous), an alert fires within 2 minutes. The system uses keyed state — a per-sensor dictionary in memory — to implement a 2-minute cooldown, preventing alert storms. The watermark ensures we use the sensor's event timestamp, not server time, so delayed messages don't trigger false alerts."

Traffic Gridlock:

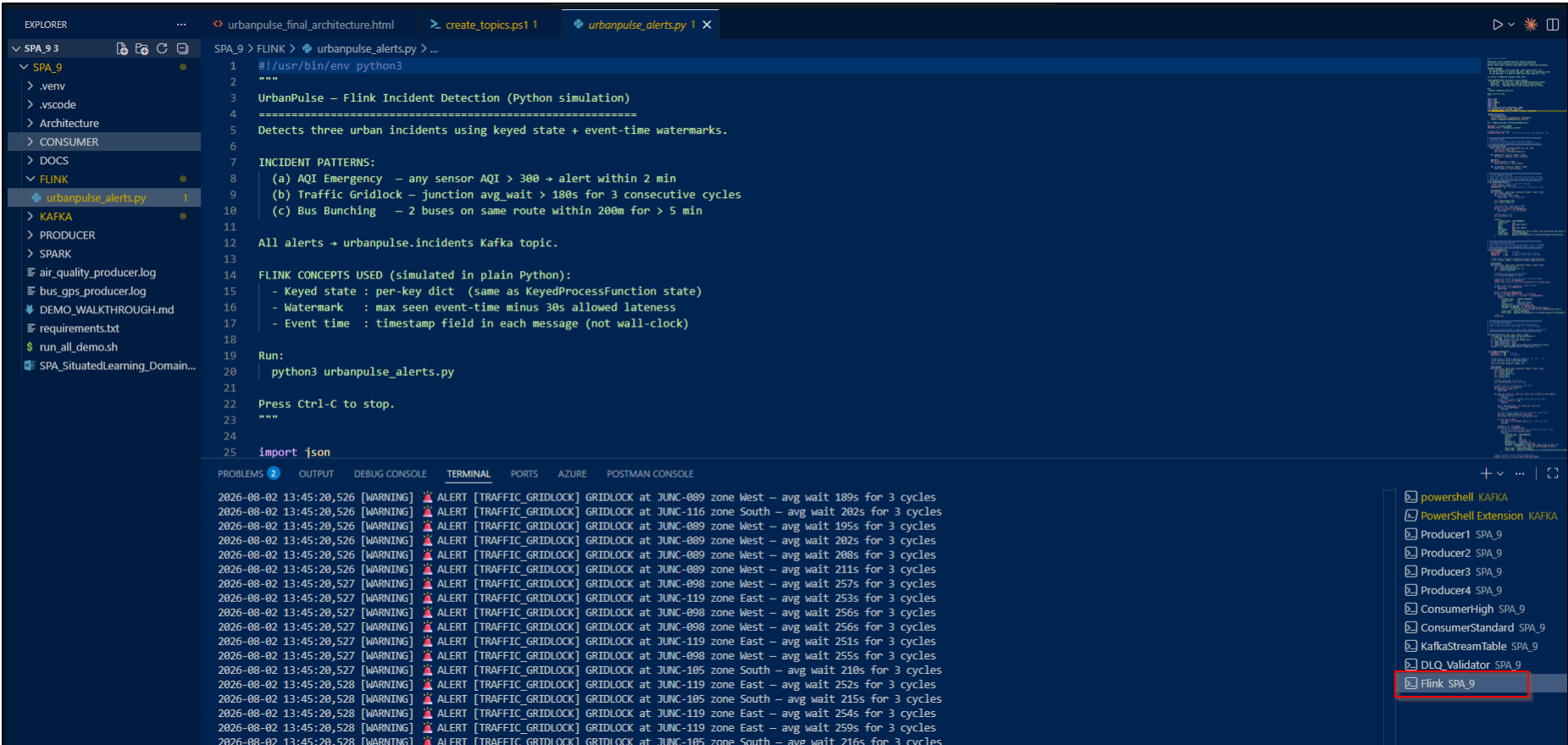
"A deque keyed by junction_id stores the last 3 avg_wait_sec readings. When all 3 consecutive readings exceed 180 seconds, a gridlock alert fires. This simulates Flink's KeyedProcessFunction with ValueState."

Bus Bunching:

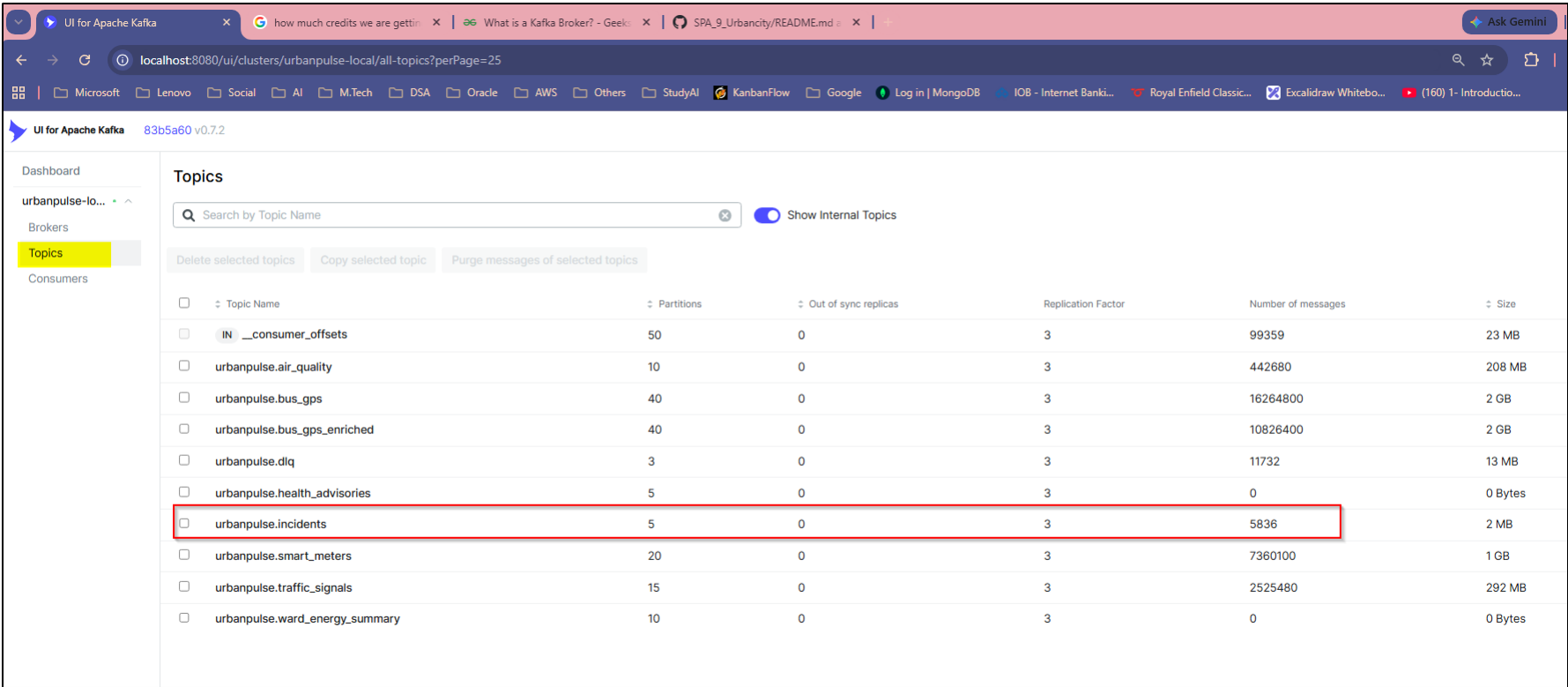
"For every pair of buses on the same route, we compute the Haversine distance — great-circle distance on Earth's surface in metres. If two buses are within 200m for more than 5 minutes, we emit a bunching alert. Keyed state per route_id stores the GPS positions."

Step 8: Run the python script for FLINK

Command: python FLINK/urbanpulse_alerts.py



We can see the Flink Alert in Topics:



← → ↺

localhost:8080/ui/clusters/urbanpulse-local/all-topics/urbanpulse.incidents

Microsoft | Lenovo | Social | AI | M.Tech | DSA | Oracle | AWS | Others | StudyAI | KanbanFlow | Google | Log in | MongoDB | IOB - Internet Banki... | Royal Enfield Classic... | Excalidraw Whitebo...

UI for Apache Kafka83b5a60 v0.7.2

Dashboard

urbanpulse-lo...

BrokersTopicsConsumers

Topics

urbanpulse.incidents

OverviewMessagesConsumersSettingsStatistics

Partitions5

Replication Factor3

URP0

In Sync Replicas15 of 15

TypeExternal

Segment Size7 MB

Segment Count15

Clean Up PolicyDELETE

Message Count8785

Partition ID

Replicas

First Offset

Next Offset

Message Count

0

2, 3, 1

0

0

0

1

3, 1, 2

0

0

0

2

1, 2, 3

0

8785

8785

3

2, 1, 3

0

0

0

4

3, 2, 1

0

0

0

- Sample message shows ‘Incident Type: AQI_Emergency’

UI for Apache Kafka83b5a60 v0.7.2

Dashboard

urbanpulse-lo...

BrokersTopicsConsumers

Topics / urbanpulse.incidents

OverviewMessagesConsumersSettingsStatistics

Seek Type

Offset

Offset

Partitions

All items are selected.

Key Serde

String

Value Serde

String

Clear all

Submit

Q AQI

+ Add Filters

DONE

1767 ms

144 MB

Offset

Partition

Timestamp

Key Preview

Value Preview

384729

2

8/2/2026, 14:26:27

AQI_EMERGENCY

{ "incident_type": "AQI_EMERGENCY", "sensor_id": "...

Key

Value

Headers

Timestamp

8/2/2026, 14:26:27

Timestamp type: CREATE_TIME

Key Serde

String

Size: 13 Bytes

Value Serde

String

Size: 271 Bytes

{ "incident_type": "AQI_EMERGENCY", "sensor_id": "AQ-037", "zone": "East", "aqi": 500, "pm25": 208.6, "threshold": 300, "message": "HAZARDOUS AQI 500 at sensor AQ-037 zone East", "event_time": "2026-08-02T08:56:27.307+00:00", "alert_time": "2026-08-02T08:56:27.330+00:00" }

+

384588

2

8/2/2026, 14:26:26

AQI_EMERGENCY

{ "incident_type": "AQI_EMERGENCY", "sensor_id": "...

+

384046

2

8/2/2026, 14:26:22

AQI_EMERGENCY

{ "incident_type": "AQI_EMERGENCY", "sensor_id": "...

+

384045

2

8/2/2026, 14:26:22

AQI_EMERGENCY

{ "incident_type": "AQI_EMERGENCY", "sensor_id": "...

+

383476

2

8/2/2026, 14:26:18

AQI_EMERGENCY

{ "incident_type": "AQI_EMERGENCY", "sensor_id": "...

+

383475

2

8/2/2026, 14:26:18

AQI_EMERGENCY

{ "incident_type": "AQI_EMERGENCY", "sensor_id": "...

+

383352

2

8/2/2026, 14:26:17

AQI_EMERGENCY

{ "incident_type": "AQI_EMERGENCY", "sensor_id": "...

+

383063

2

8/2/2026, 14:26:15

AQI_EMERGENCY

{ "incident_type": "AQI_EMERGENCY", "sensor_id": "...

- Sample message shows ‘Incident Type: TRAFFIC_GRIDLOCK’

Topics / urbanpulse.incidents

OverviewMessagesConsumersSettingsStatistics

Seek Type

Offset

Offset

Partitions

All items are selected.

Key Serde

String

Value Serde

String

Clear all

Cancel

Q BUS

+ Add Filters

Polling partitions: [2]

4424 ms

206 MB

Offset

Partition

Timestamp

Key Preview

Value Preview

372416

2

8/2/2026, 14:25:02

BUS_BUNCHING

{ "incident_type": "BUS_BUNCHING", "route_id": "R...

Key

Value

Headers

Timestamp

8/2/2026, 14:25:02

Timestamp type: CREATE_TIME

Key Serde

String

Size: 12 Bytes

Value Serde

String

Size: 321 Bytes

{ "incident_type": "BUS_BUNCHING", "route_id": "R006", "bus_1": "BUS-0205", "bus_2": "BUS-0480", "distance_m": 197.2, "duration_sec": 300.0, "message": "BUNCHING on route R006: buses BUS-0205 & BUS-0480 within 197m for 5.0 min", "event_time": "2026-08-02T08:55:02.861+00:00", "alert_time": "2026-08-02T08:55:02.910+00:00" }

+

368268

2

8/2/2026, 14:24:33

BUS_BUNCHING

{ "incident_type": "BUS_BUNCHING", "route_id": "R...

+

327117

2

8/2/2026, 14:20:02

BUS_BUNCHING

{ "incident_type": "BUS_BUNCHING", "route_id": "R...

+

303942

2

8/2/2026, 14:17:37

BUS_BUNCHING

{ "incident_type": "BUS_BUNCHING", "route_id": "R...

+

280159

2

8/2/2026, 14:15:02

BUS_BUNCHING

{ "incident_type": "BUS_BUNCHING", "route_id": "R...

+

276598

2

8/2/2026, 14:14:36

BUS_BUNCHING

{ "incident_type": "BUS_BUNCHING", "route_id": "R...

+

247518

2

8/2/2026, 14:11:09

BUS_BUNCHING

{ "incident_type": "BUS_BUNCHING", "route_id": "R...

- Sample message shows ‘Incident Type: **BUS_BUNCHING**’

Topics / urbanpulse.incidents

Overview

Messages

Consumers

Settings

Statistics

Seek Type

Offset

Offset

All items are selected.

Key Serde

String

Value Serde

String

Clear all

Cancel

Q BUS

+ Add Filters

Offset

Partition

Timestamp

Key

Value

Preview

372416

2

8/2/2026, 14:25:02

BUS_BUNCHING

{("incident_type": "BUS_BUNCHING", "route_id": "R...

Timestamp

8/2/2026, 14:25:02

Timestamp type: CREATE_TIME

Key Serde

String

Size: 12 Bytes

Value Serde

String

Size: 321 Bytes

Key

Value

Headers

{
 "incident_type": "BUS_BUNCHING",
 "route_id": "R006",
 "bus_1": "BUS-0205",
 "bus_2": "BUS-0480",
 "distance_m": 197.2,
 "duration_sec": 300.0,
 "message": "BUNCHING on route R006: buses BUS-0205 & BUS-0480 within 197m for 5.0 min",
 "event_time": "2026-08-02T08:55:02.861+00:00",
 "alert_time": "2026-08-02T08:55:02.910+00:00"
}

368268

2

8/2/2026, 14:24:33

BUS_BUNCHING

{("incident_type": "BUS_BUNCHING", "route_id": "R...

327117

2

8/2/2026, 14:20:02

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8/2/2026, 14:14:36

BUS_BUNCHING

{("incident_type": "BUS_BUNCHING", "route_id": "R...

247518

2

8/2/2026, 14:11:09

BUS_BUNCHING

{("incident_type": "BUS_BUNCHING", "route_id": "R...

We also noticed the **Flink Incident** in **Consumers**:

UI for Apache Kafka

83b5a80 v0.7.2

Dashboard

urbanpulse-lo...

Brokers

Topics

Consumers

Consumers

Search by Consumer Group ID

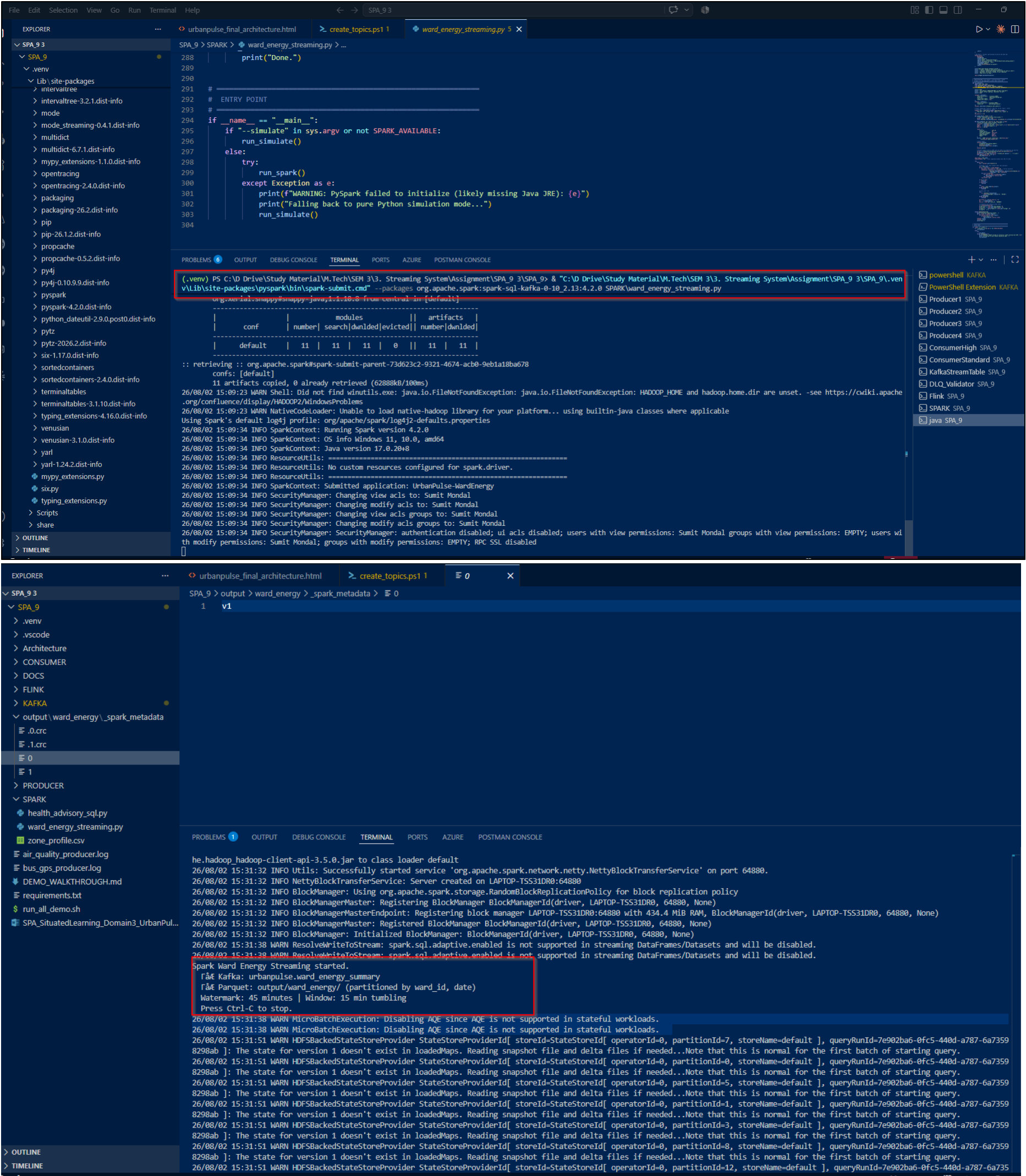
Group ID	Num Of Members	Num Of Topics	Consumer Lag	Coordinator	State
HIGH_PRIORITY_GROUP	1	1	760	3	STABLE
STANDARD_PRIORITY_GROUP	1	1	2209803	1	STABLE
bus-gps-enrichment-job	1	1	4800	1	STABLE
dlq-validator	1	4	15760	1	STABLE
flink-incident-detector	1	3	8520	3	STABLE

Problem Statements — Part II: Spark Structured Streaming (Ward Analytics)

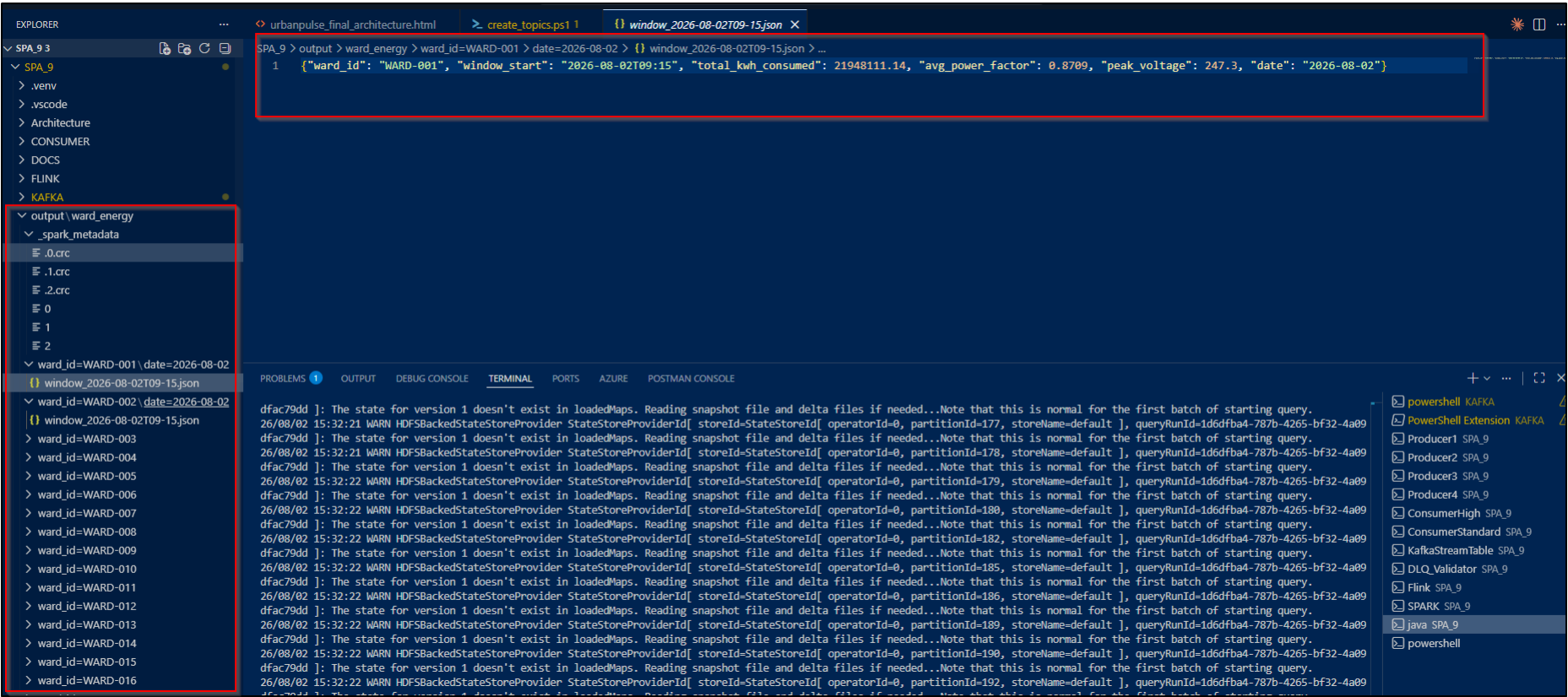
Step 9: Run the python script for ward_energy_streaming.py

Ward Energy Analytics (ward_energy_streaming.py): Consumes smart meter readings from Kafka using 15-minute tumbling windows and a 45-minute watermark to calculate total kWh consumption, average power factor, and peak voltage per ward, writing the aggregated metrics to both a Kafka summary topic and partitioned Parquet storage

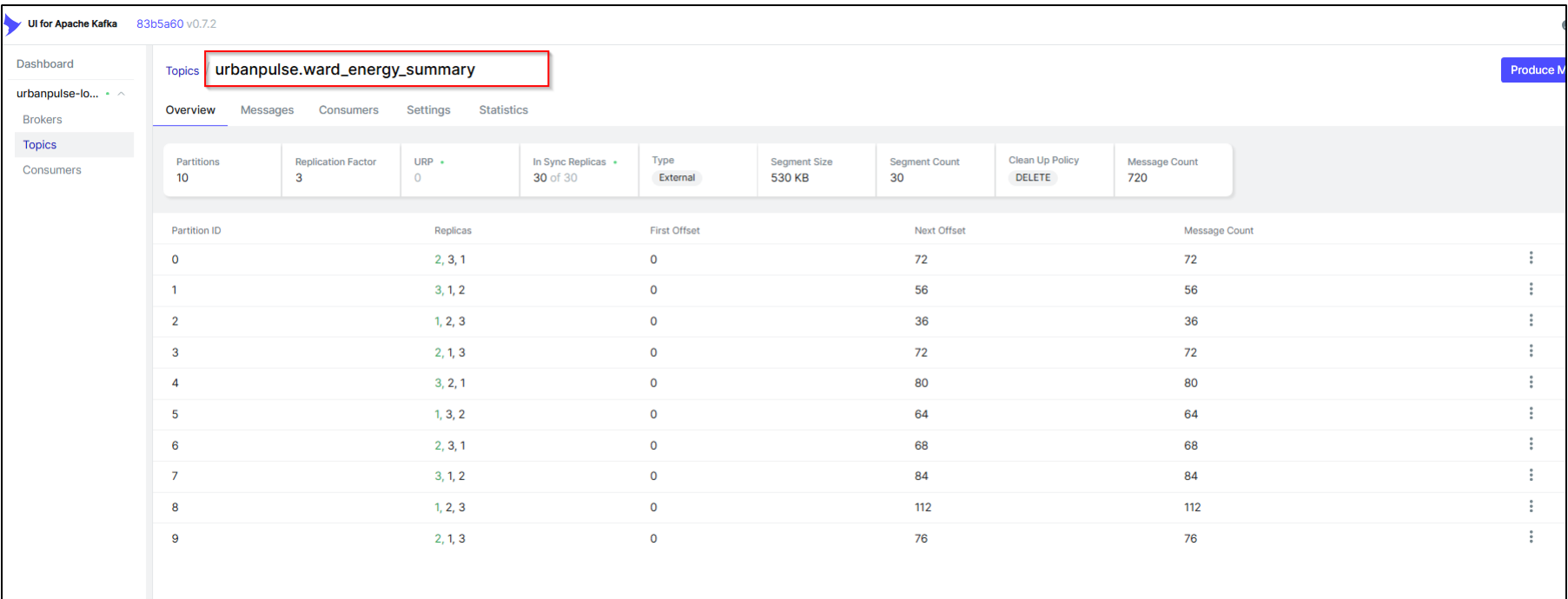
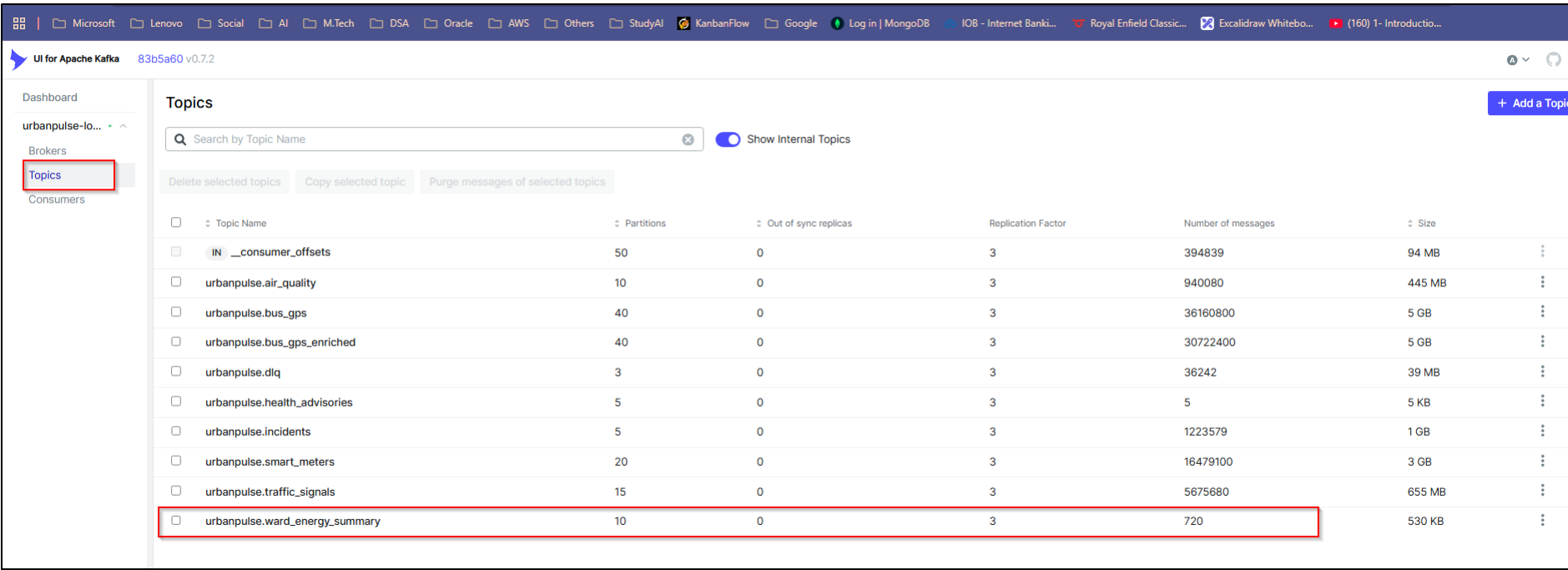
Command: python SPARK/ward_energy_streaming.py



- It has created the `output\ward_energy`: (Spark Structured Streaming application on urbanpulse.smart_meters computing, per ward_id, per 15-minute tumbling window)



- We can also see the same output in `APACHE KAFKA topics`:



UI for Apache Kafka83b5a60 v0.7.2

Dashboard

urbanpulse-lo...

Brokers

Topics

Consumers

Topics / urbanpulse.ward_energy_summary

OverviewMessagesConsumersSettingsStatistics

Seek Type

Offset

Offset

Partitions

All items are selected.

Key Serde

String

Value Serde

String

Clear all

Submit

Q Search

+ Add Filters

DONE

12 ms

138

	Offset	Partition	Timestamp	Key Preview	Value Preview
	0	9	8/2/2026, 14:45:27	WARD-046	{ "ward_id": "WARD-046", "window_start": "2026-...
	0	6	8/2/2026, 14:45:27	WARD-028	{ "ward_id": "WARD-028", "window_start": "2026-...

KeyValueHeaders

{ "ward_id": "WARD-028", "window_start": "2026-08-02T09:00", "total_kwh_consumed": 46520167.58, "avg_power_factor": 0.9092, "peak_voltage": 250.0, "date": "2026-08-02" }

Timestamp

8/2/2026, 14:45:27

Timestamp type: CREATE_TIME

Key Serde

String

Size: 8 Bytes

Value Serde

String

Size: 167 Bytes

- We also noticed the ward energy in Consumers:

UI for Apache Kafka83b5a60 v0.7.2

Dashboard

urbanpulse-lo...

Brokers

Topics

Consumers

Consumers

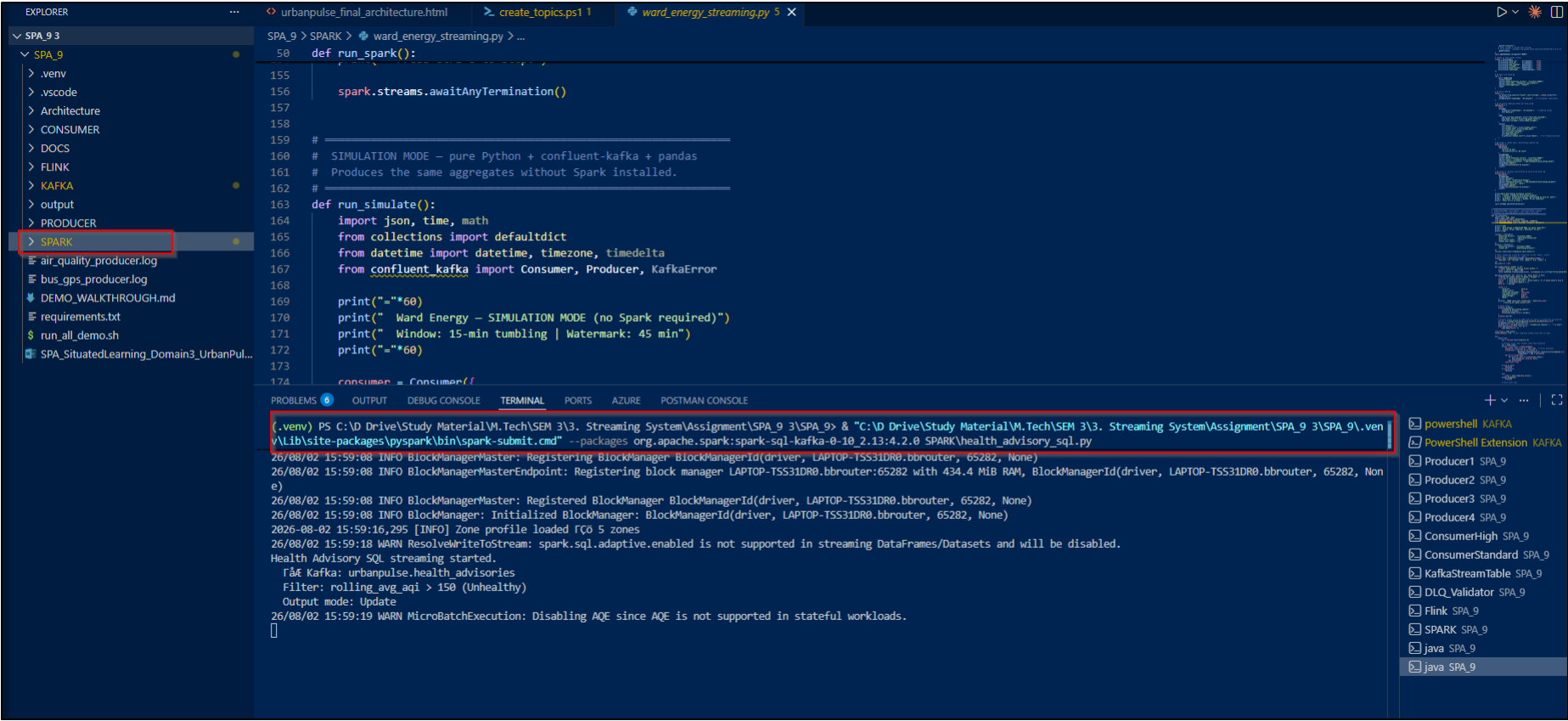
Q Search by Consumer Group ID

Group ID	Num Of Members	Num Of Topics	Consumer Lag	Coordinator	State
HIGH_PRIORITY_GROUP	1	1	1410	3	STABLE
STANDARD_PRIORITY_GROUP	1	1	5329081	1	STABLE
bus-gps-enrichment-job	1	1	11687	1	STABLE
dlq-validator	1	4	3622	1	STABLE
flink-incident-detector	1	3	14200	3	STABLE
health-advisory-sql-sim	0	1	46260	1	EMPTY
spark-ward-energy-sim	4	1	2772	2	STABLE

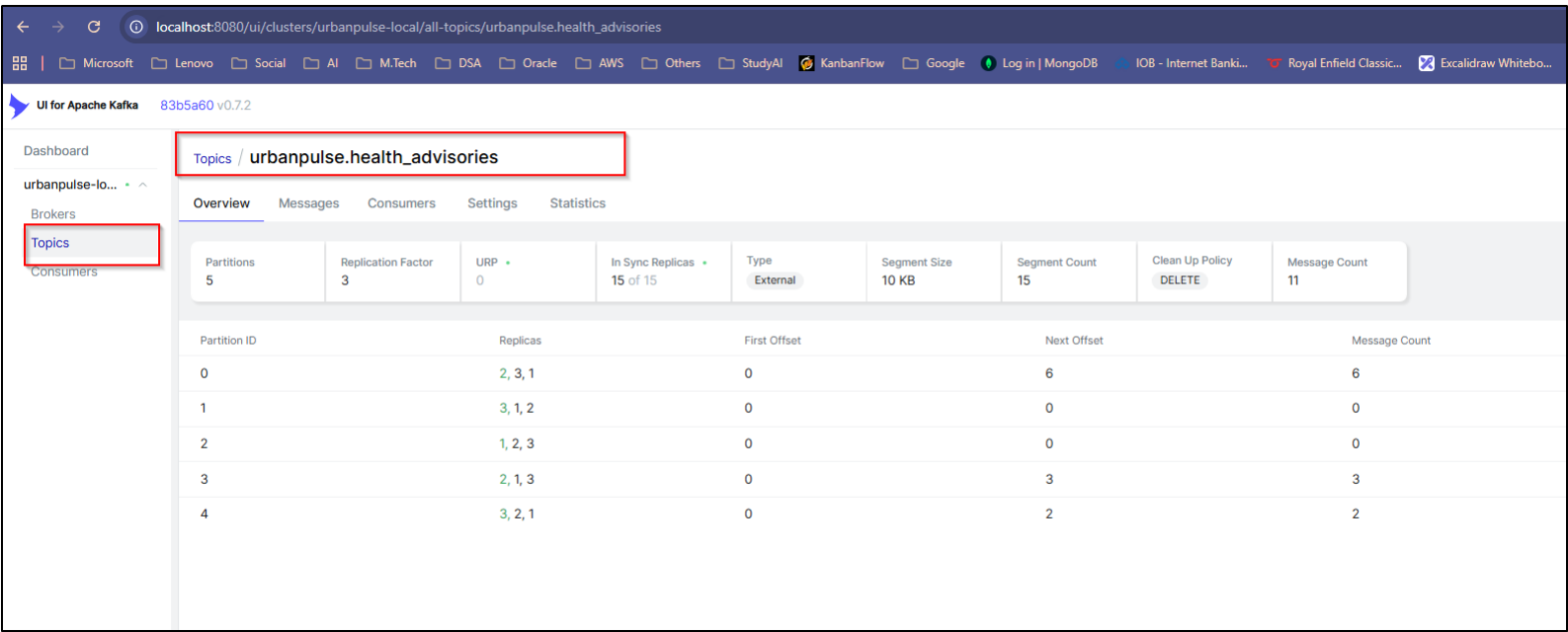
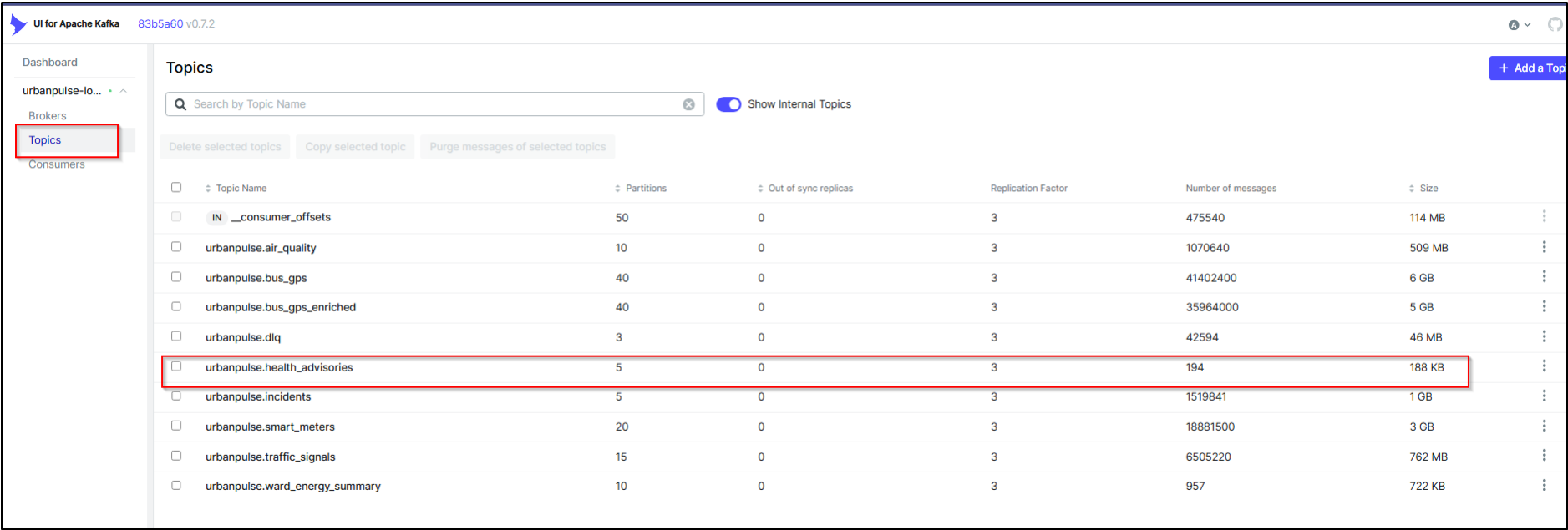
Step 10: Run the python script for ward_energy_streaming.py

Health Advisory Generator (health_advisory_sql.py): Processes streaming air quality data from Kafka to compute 10-minute rolling average AQI scores per zone, enriches this data with static zone profile metadata (such as population, schools, and hospitals), filters for unhealthy AQI levels (> 150), and streams health advisories back to Kafka

Command: python SPARK/health_advisory_sql.py



- We can also see the same output in **APACHE KAFKA Topics**:



UI for Apache Kafka83b5a60 v0.7.2

Dashboard

urbanpulse-lo...

Brokers

Topics

Consumers

Topics / urbanpulse.health_advisories

Overview

Messages

Consumers

Settings

Statistics

Seek Type

Offset

Offset

Partitions

All items are selected.

Key Serde

String

Value Serde

String

Clear all

Submit

Search

+ Add Filters

DONE

2 ms

	Offset	Partition	Timestamp	Key	Value
+	0	4	8/2/2026, 16:00:15	South	{ "zone": "South", "zone_name": "South MetroConne...
+	0	0	8/2/2026, 16:00:15	East	{ "zone": "East", "zone_name": "East MetroConnect", ...
-	0	3	8/2/2026, 16:00:15	Central	{ "zone": "Central", "zone_name": "Central MetroCon...

Key

Value

Headers

```
{
  "zone": "Central",
  "zone_name": "Central MetroConnect",
  "window_start": "2026-08-02T15:50:00.000+05:30",
  "window_end": "2026-08-02T16:00:00.000+05:30",
  "rolling_avg_aqi": 262.26785714285717,
  "population": 195000,
  "num_schools": 18,
  "num_hospitals": 8,
  "sensor_readings": 616
}
```

Timestamp

8/2/2026, 16:00:15

Timestamp type: CREATE_TIME

Key Serde

String

Size: 7 Bytes

Value Serde

String

Size: 259 Bytes